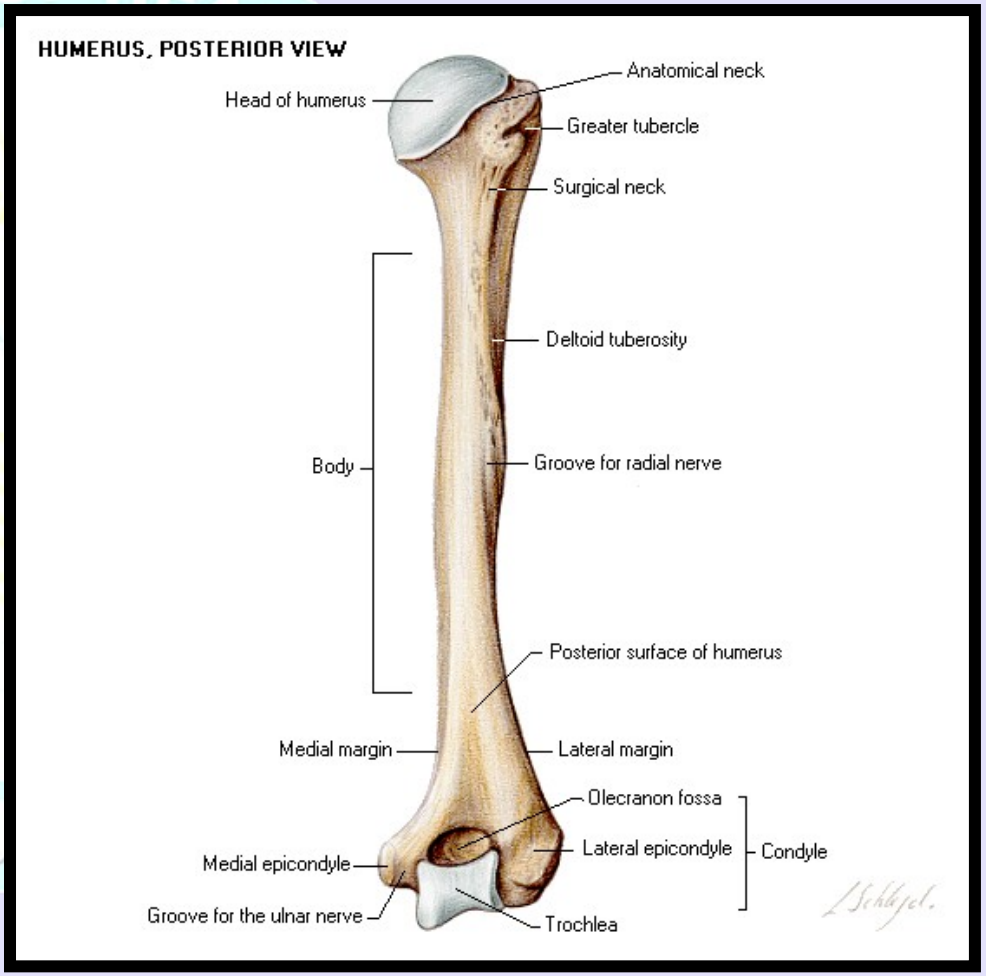
# CLASSIFICATION & HISTOLOGY OF BONE

# LEARNING OBJECTIVES

- At the end of the lecture the student should be able to:
  - Recognize bone and its functions and composition.
  - Differentiate between woven bone and lamellar bone.
  - Differentiate between compact bone and spongy bone.
  - Learn the applied aspect of bone

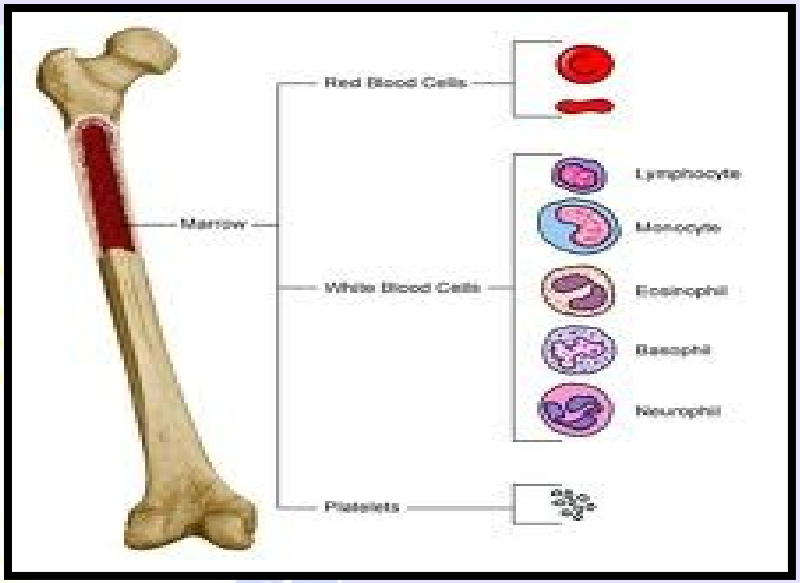
# BONE

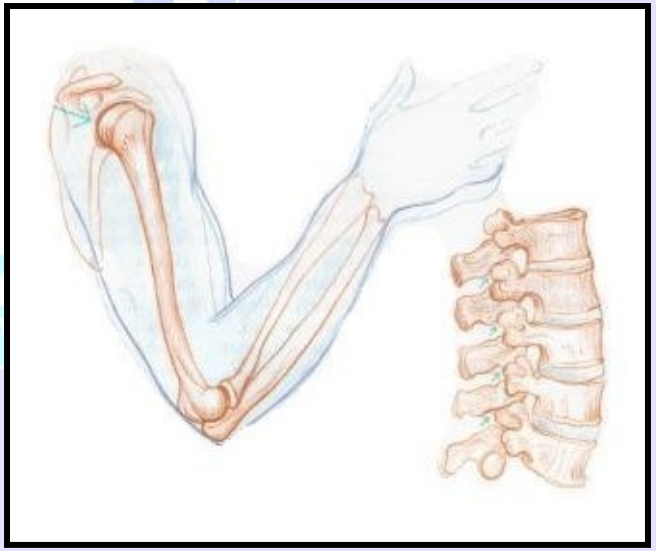
- **Bone** is a specialized type of connective tissue characterized by intercellular calcified material - **The bone matrix**



# FUNCTIONS OF BONE

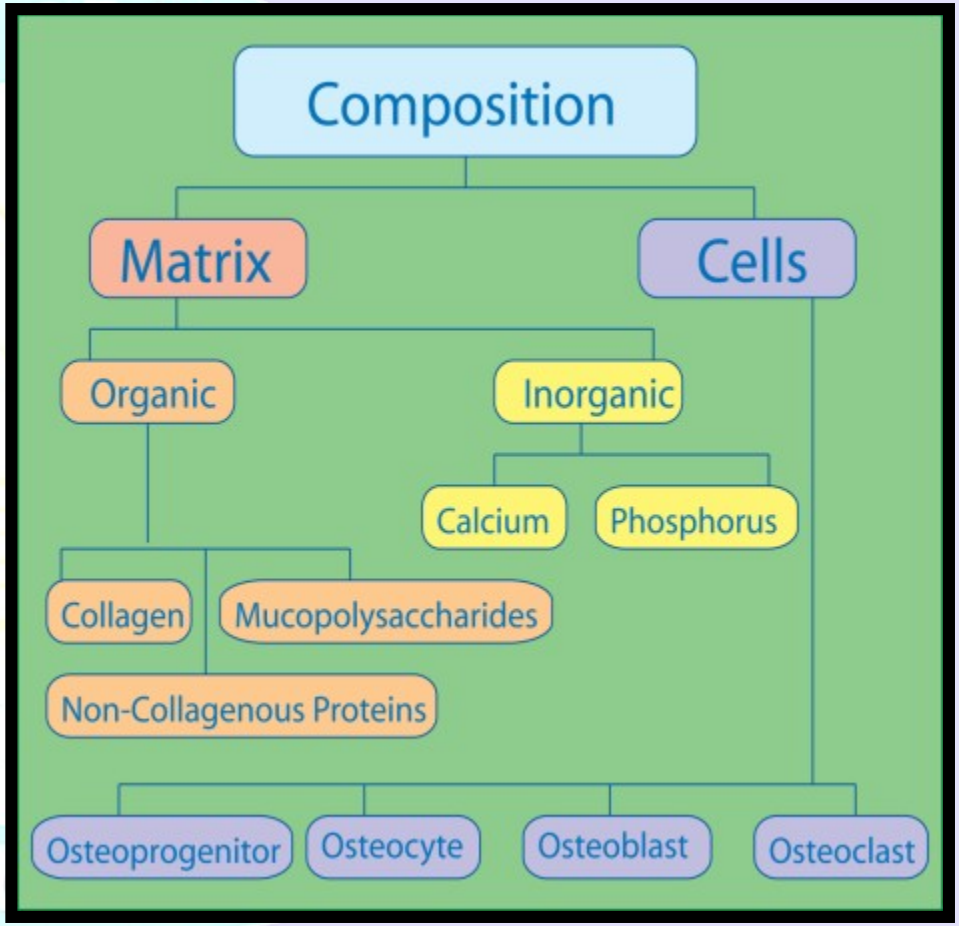
- Supports soft tissue
- Protects vital organs (cranium, thoracic cavity)
- Contains bone marrow
- Reservoir of Ca, PO4 and maintains their constant concentrations in body fluids
- Locomotion



# COMPOSITION

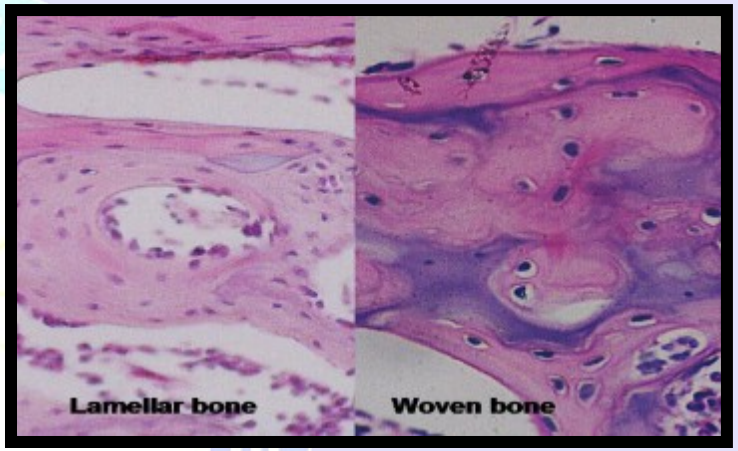
- **Cells**
  - Osteoblasts
  - Osteocytes
  - Osteoclasts
- **Bone matrix**
  - Organic:
    Fiber + ground substance (Proteoglycan and glycoprotein)
  - Inorganic:
    Calcium + phosphorus (hydroxyapatite crystal & noncrystalline amorphous form), bicarbonates and citrate etc



# TYPES OF BONE

**Microscopically**

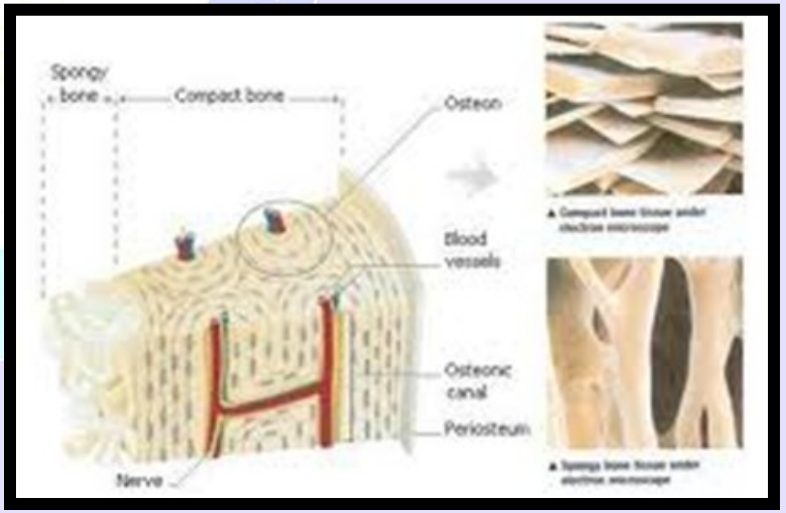
- Primary ( woven)
- Secondary (Lamellar)




**Macroscopically**

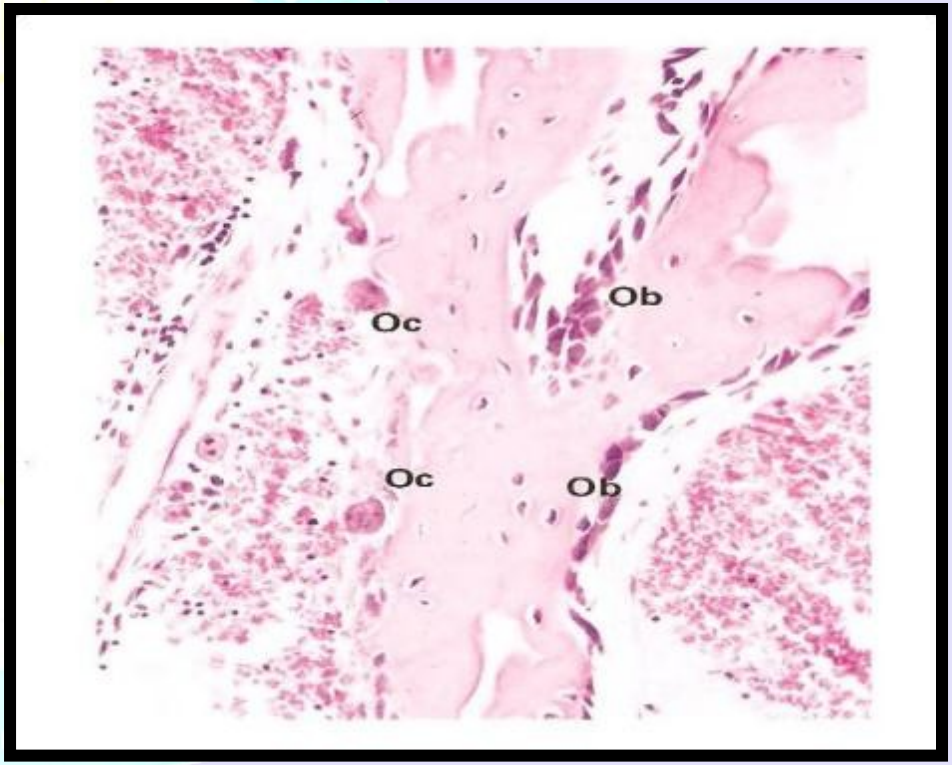
- Compact
- Cancellous

(Both have lamellar arrangement)



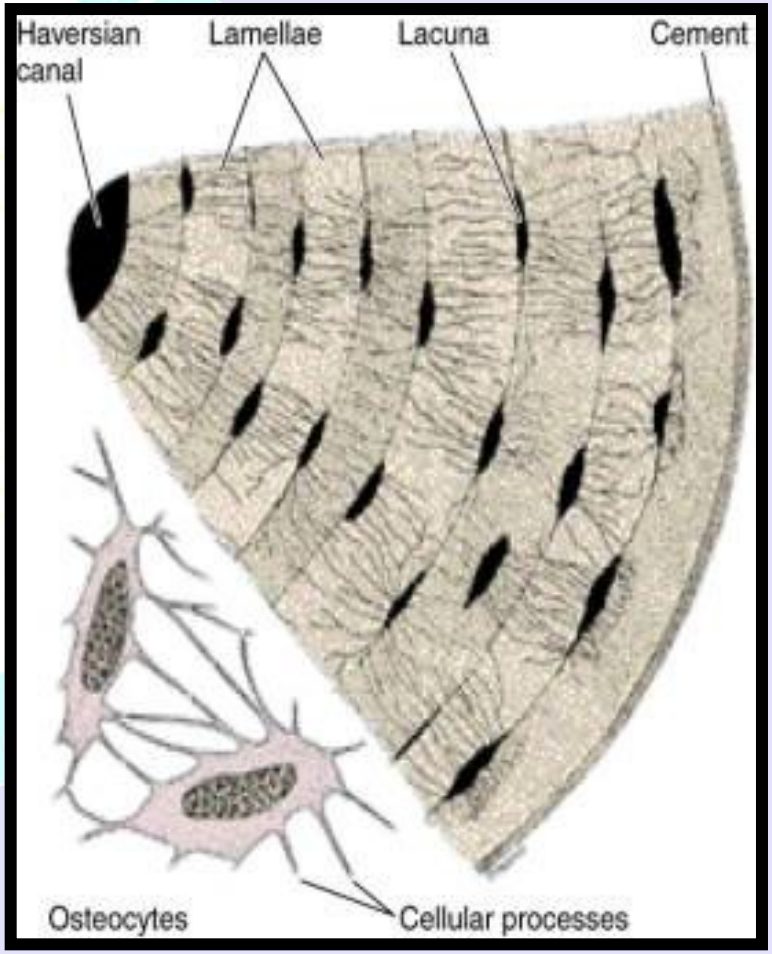
# WOVEN BONE

- Characterized by random deposition of fine collagen and increased cellularity.
- Relatively radiolucent (low mineral contents).
- Seen in embryonic development, fracture and repair.



# LAMELLAR BONE

- Characterized by
- Arrangement of collagen fibers in lamellae organized concentrically around a vascular canal containing blood vessels and nerves.
- The lamellae and the vascular canal constitute the Haversian system or osteon.
- Lacunae containing osteocytes are present between the lamellae.
- Mineralized matrix with few fibers present around each osteon, called the cementing substance.

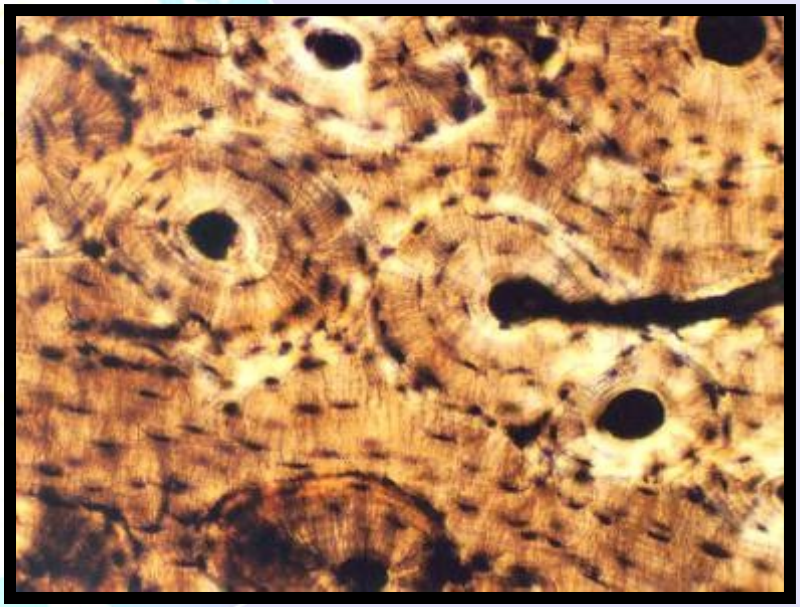


# DIFFERENCE BETWEEN WOVEN AND LAMELLAR BONES

- Woven (primary) bone tissue is temporary and is replaced in adults by Lamellar (secondary) bone tissue except in few places i.e. near sutures of flat bones of skull
- There is irregular array of collagen fibers in Woven (primary) bone tissue
- Woven (primary) bone tissue has lower mineral content
- Woven (primary) bone tissue has higher proportion of osteocytes

# COMPACT BONE

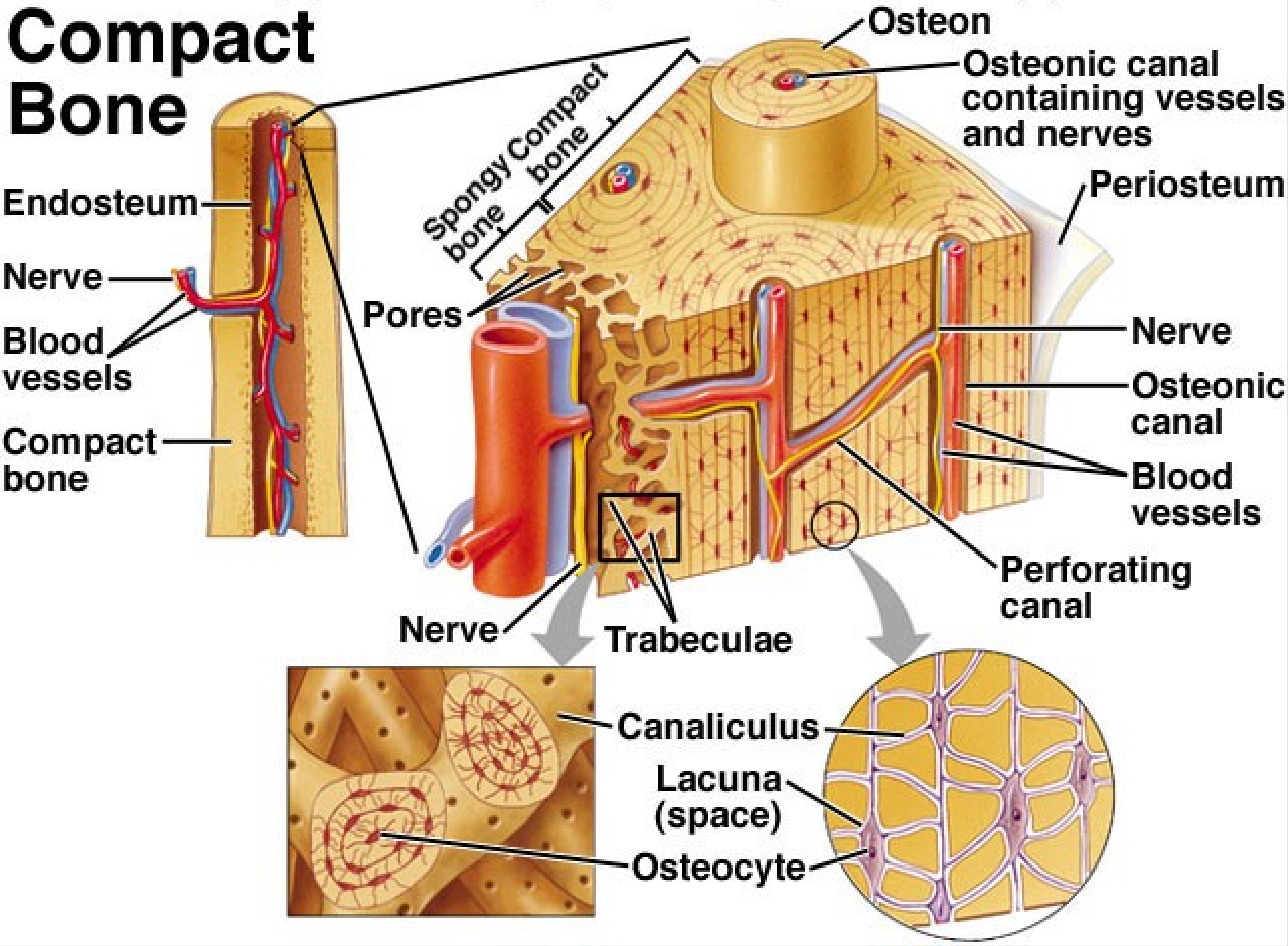
- 4 Types of lamellae seen:
  - a. Haversian system
  - b. Outer circumferential
  - c. Inner circumferential
  - d. Interstitial



- Volkman's canal communicate the osteons and, periosteal and endosteal surfaces

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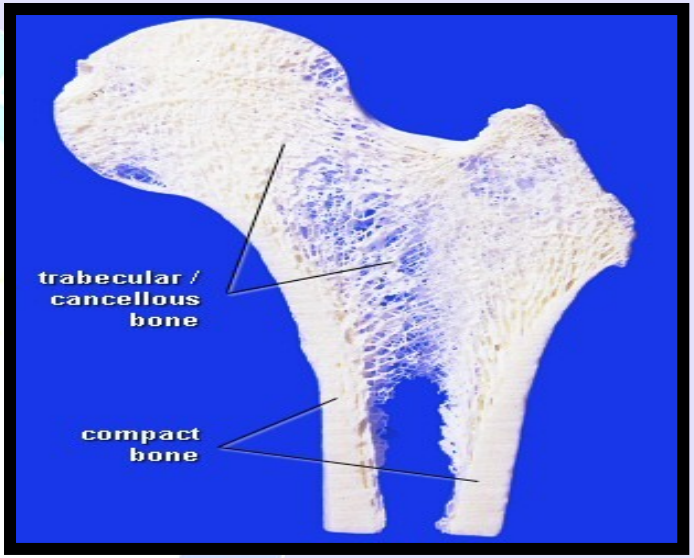
# Compact Bone



# SPONGY BONE

- Bony tissue is in the form of trabeculae having lamellar arrangement.

- Vascular canals are absent as the vessels in marrow cavities provide nutrition by means of diffusion.



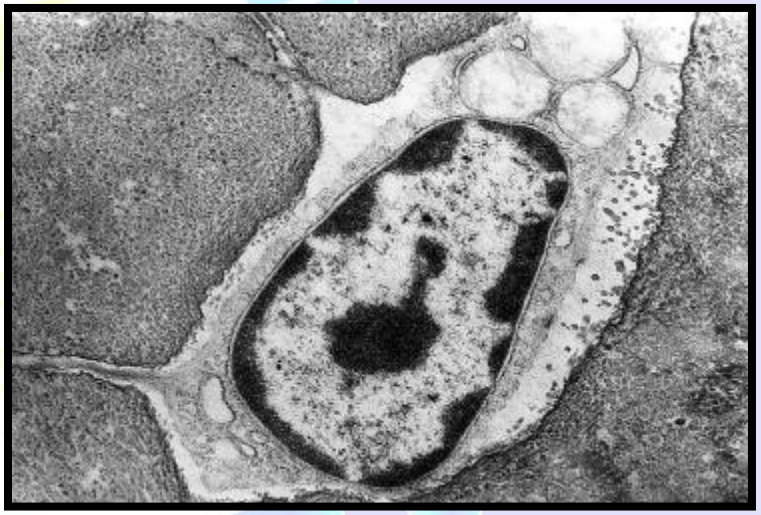



# OSTEOBLASTS

- Synthesize organic components of matrix (collagen type 1, proteoglycans, glycoproteins.)
- Epitheloid appearance, cuboidal or columnar shape
- Shows alkaline phosphatase activity
- Influence deposit of Ca+ +, PO4.
- Estrogen, PTH stimulate its activity

# OSTEOCYTES

- Mature bone cells that sit in lacunae
- Almond shaped cell with condensed nuclear chromatin
- Gap junctions between osteocytes provide nutrition Maintain bony matrix; death causes resorption of matrix
- Stimulated by calcitonin; inhibited by PTH

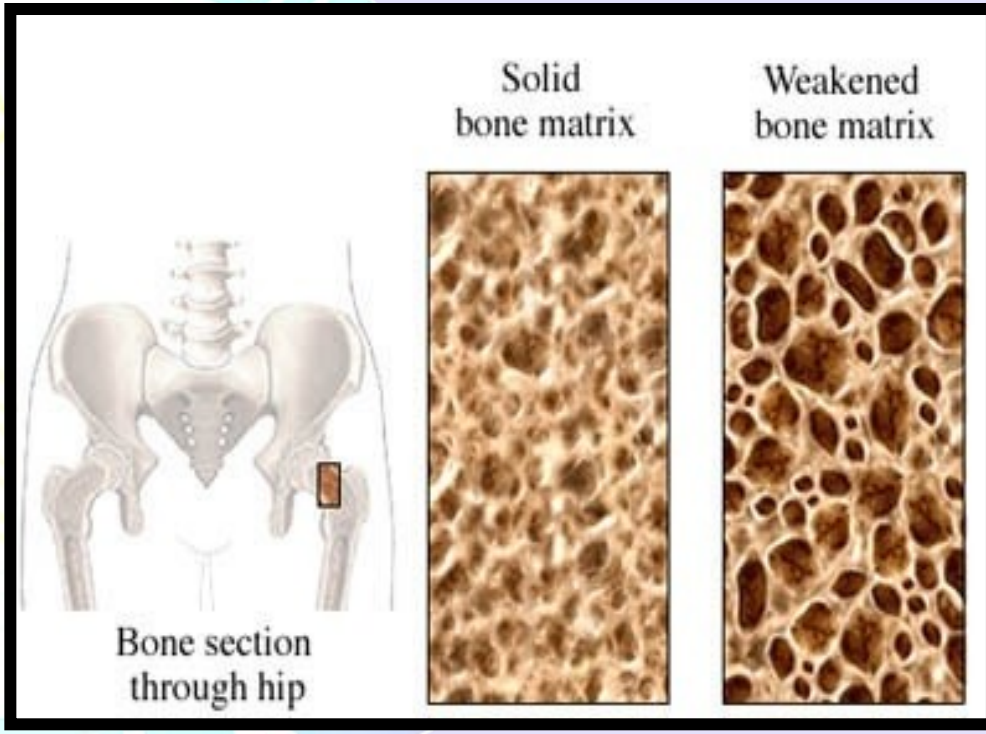


# OSTEOCLASTS

- Derived from monocytes; engulf bony material
- Large, multinucleated and branched found in Howship's lacunae
- Surface foldings forming ruffled border increasing the active resorptive area.
- Secrete enzymes (colleganase etc) that digest matrix

# BONE MATRIX

- Organic – 35% collagen
  - Chondroitin sulfate, hyaluronic acid and osteocalcin
- Inorganic – 65% calcium phosphate minerals – hydroxyapatite crystals, which binds water around it. This hydration shell facilitate exchange of ions between it and body fluids.
- Combination provides strength and rigidity. Laid down by osteoblasts.



## PERIOSTEUM

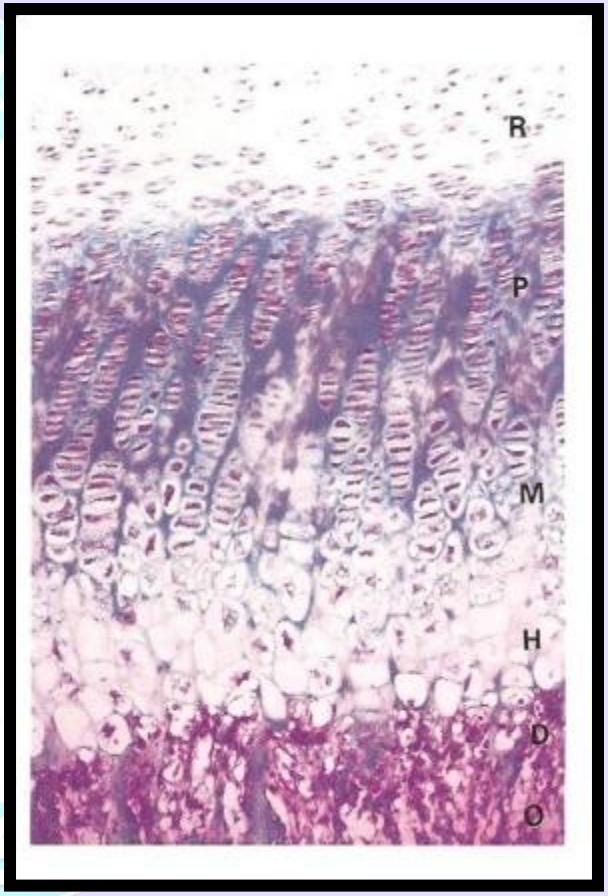
- Consists of outer layer of collagen fibers and fibroblast. The fibers penetrate the bone matrix the **Sharpey's fiber**
- Inner layer of reserve cells which can differentiate in osteoblast cells

## ENDOSTEUM

- Very little connective tissue and single layer of reserve cells
- Lines medullary and marrow cavities of bone

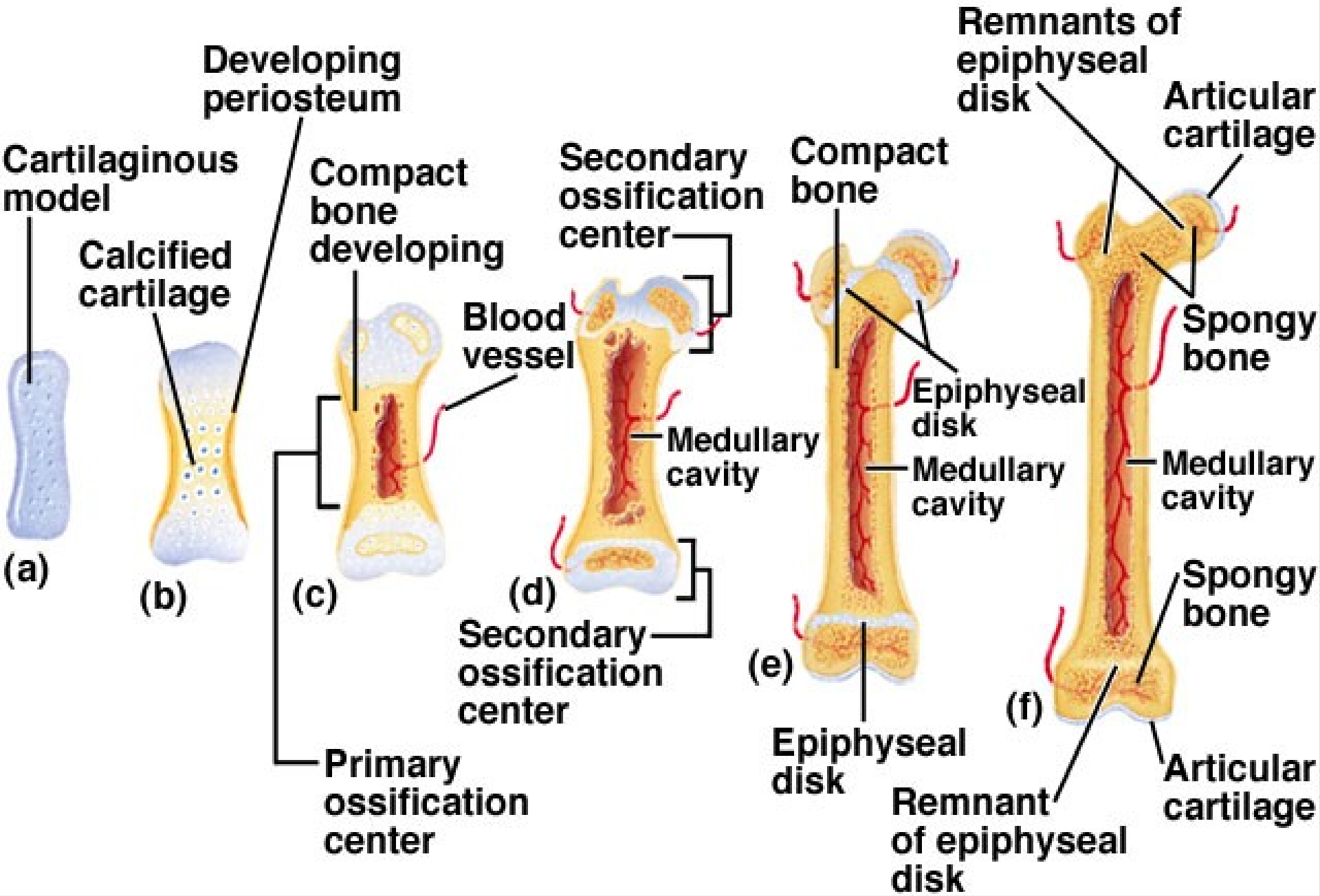
# ENDOCHONDRIAL BONE FORMATION

- Within the model of hyaline cartilage.
- Hypertrophy and destruction of chondrocytes.
- Calcification of cartilage matrix.
- Arrival of osteogenic bud.
- Osteoblast differentiate from progenetor cells and form bone matrix.



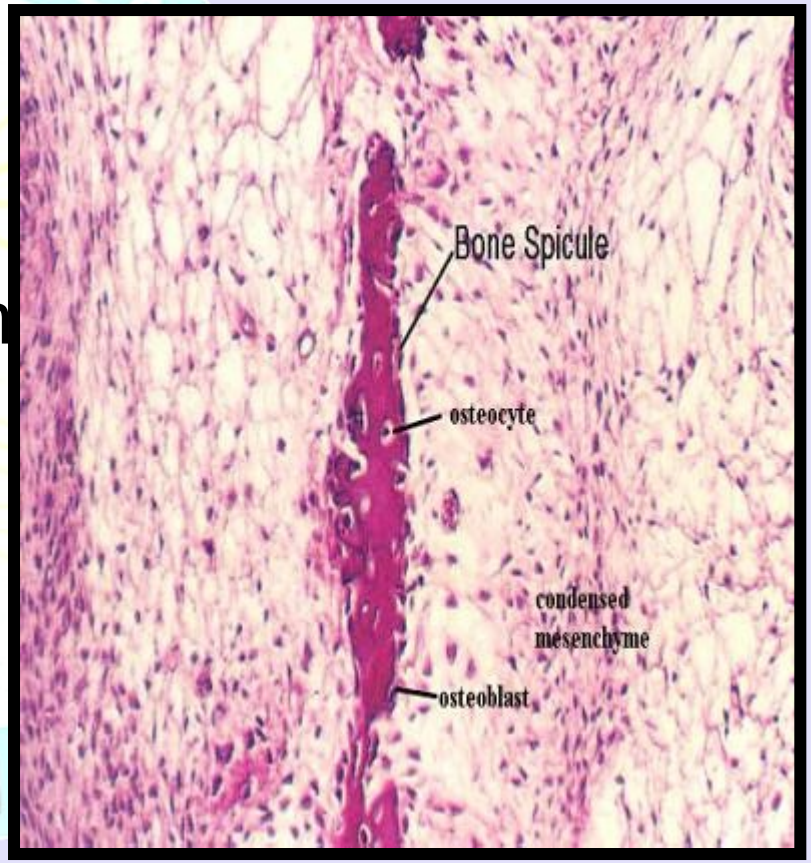

Epiphysial growth plate

# Endochondral Bone Development



# INTRAMEMBRANOUS BONE FORMATION

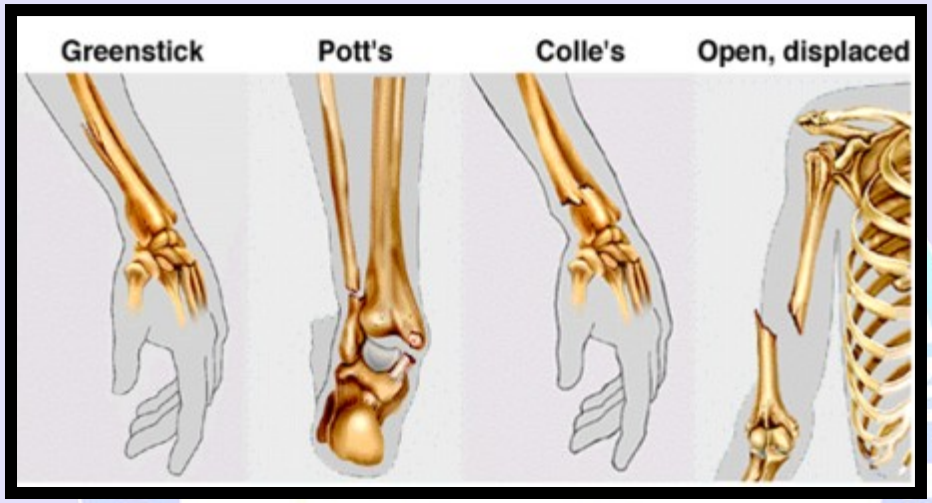
- Connective tissue develops embryonically and gives rise to bone.
- Cells in connective tissue differentiate into osteoblasts.
- Osteoblasts lay down collagen eventually starts to calcify and become osteocytes.
- Cancellous bone eventually becomes compact bone.

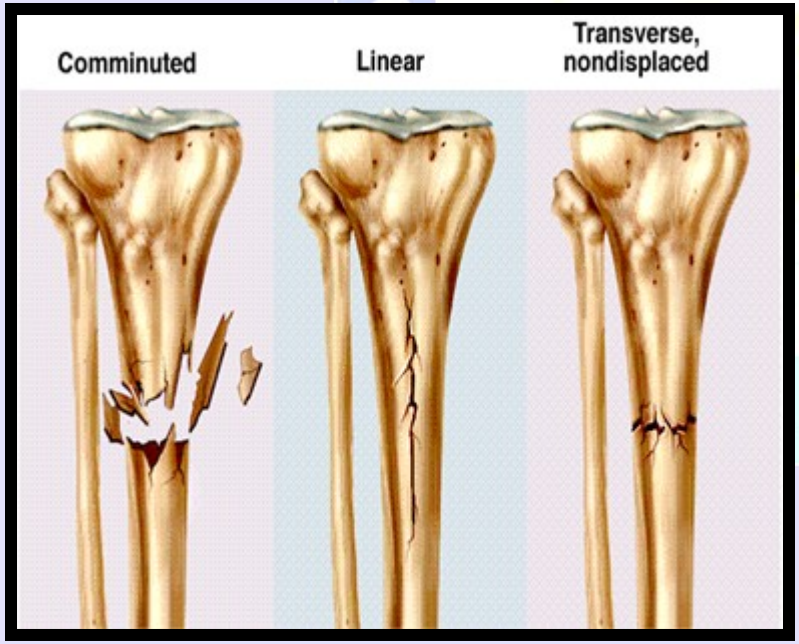


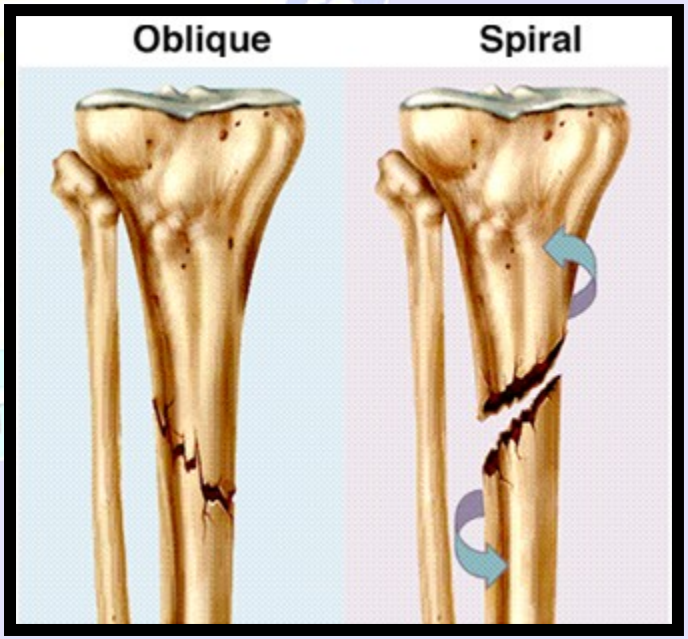
# APPLIED ANATOMY

- **Fractures**
  - **Open and**
  - **Closed**
- **Healing of these takes place in three stages;**
  - **repair by granulation tissue**
  - **union by callus**
  - **consolidation by mature bone**

# TYPES OF FRACTURES

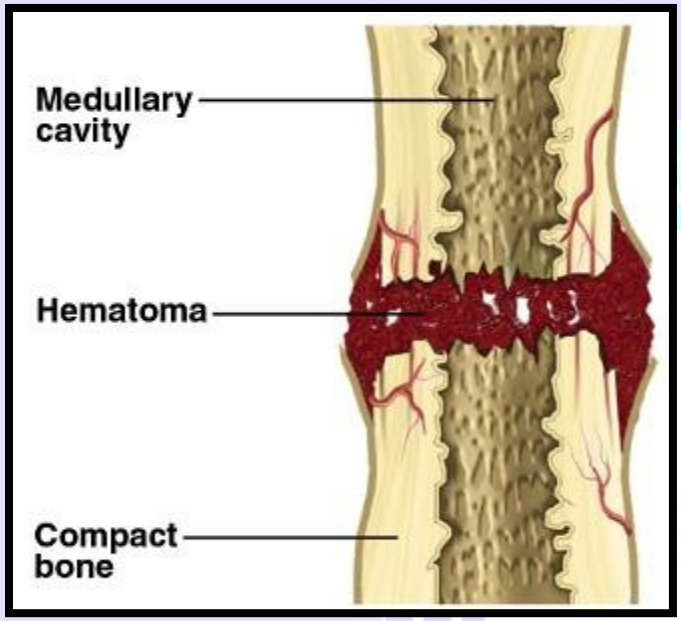


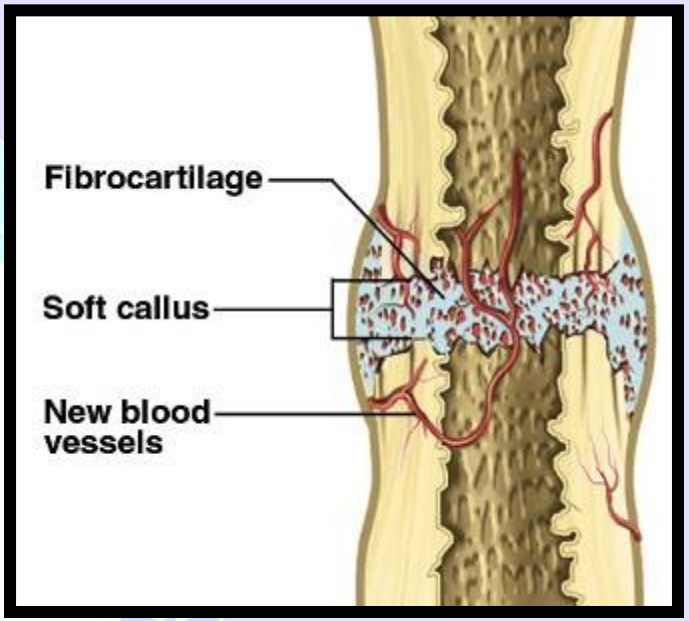



# HEALING OF FRACTURES

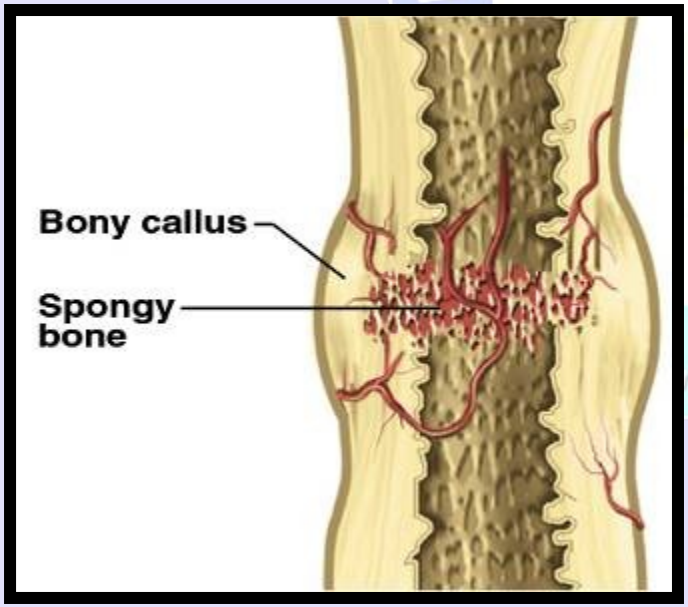
1



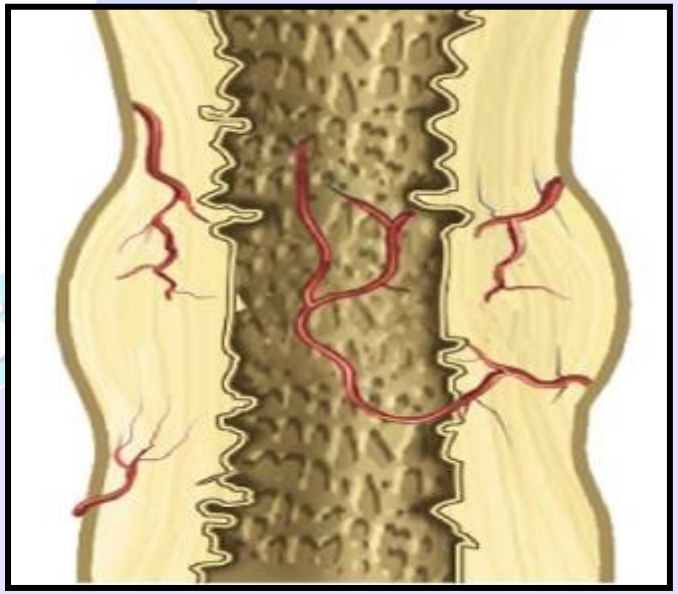

2



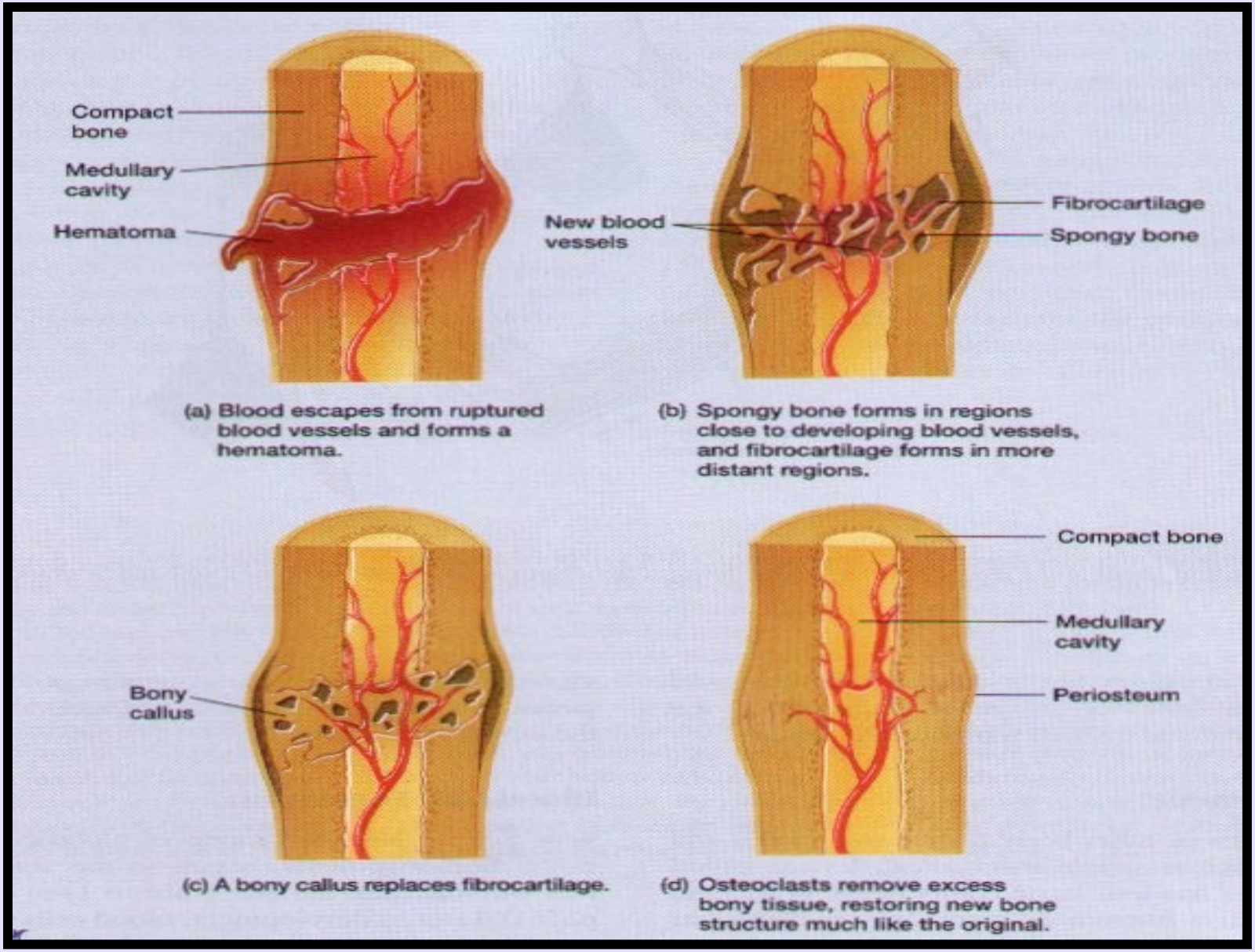

3




4



# FRACTURE REPAIR

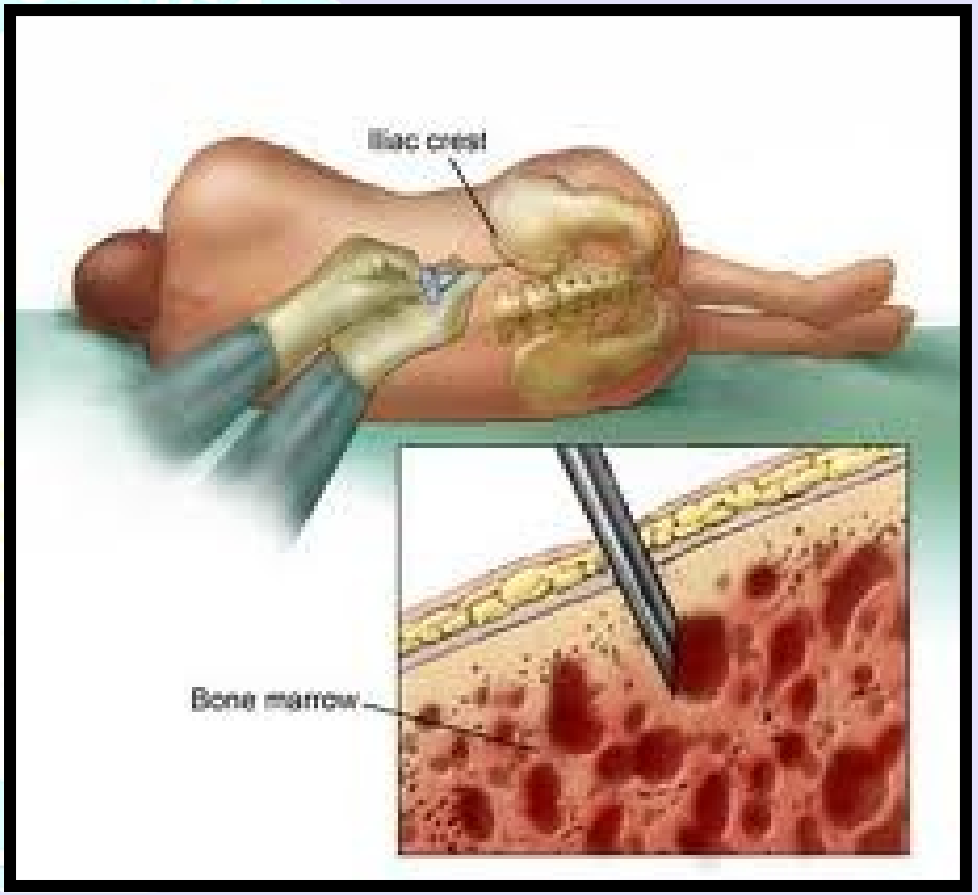


# OSTEOPOROSIS

- Most common bone disease
- Bones lose mass & becomes brittle due to loss of both organic matrix & minerals
  - Risk of fracture of hip, wrist & vertebral column.
  - Leads to fatal complications such as pneumonia or pulmonary embolism.
  - Widow's (dowager's) hump is deformed spine
  - Postmenopausal white women at greatest risk
    - By age 70, average loss is 30% of bone mass
    - ERT slows its progress, but best treatment is prevention -- exercise & calcium intake (1000 mg/day) between ages 25 and 40.

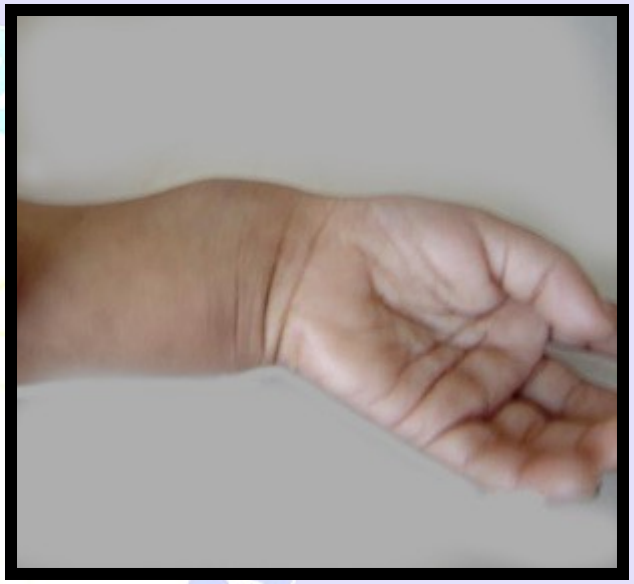
# BONE MAROW BIOPSY

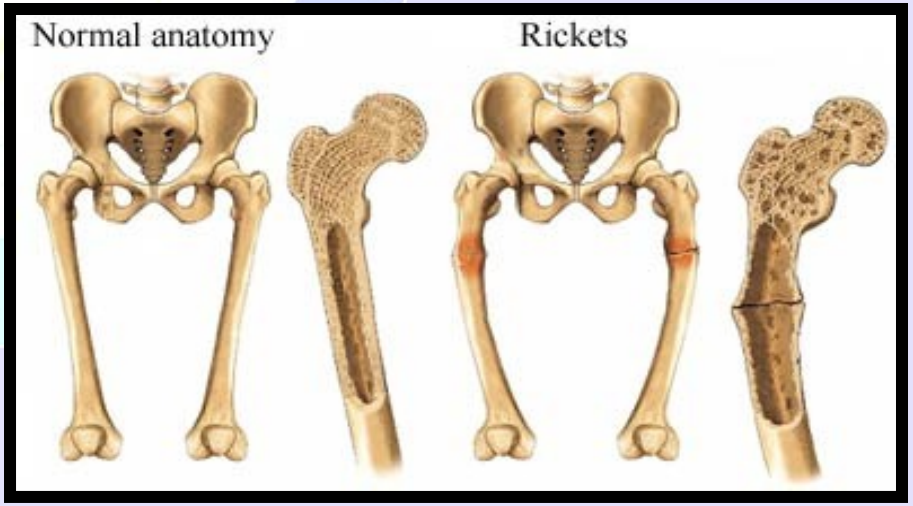
- Passing a metal pin into the medullary cavity, during bone marrow biopsy hardly interferes with the blood supply of the bone, as it is richly supplied.



# RICKETS

- Calcification of cartilage fails and ossification of the growth zone is disturbed.
- Osteoid tissue is formed normally and the cartilage cells proliferate freely, but mineralization does not take place.
- This results in craniotabs, rachitic rosary, enlarged epiphyses in limb bones, and spinal & pelvic deformities.
  -Fractures, tumors, and infections of the bone are very painful.





# REFERENCES


- MEDICAL HISTOLOGY BY LAIQ HUSSAIN SIDDIQUI
- BASIC HISTOLOGY BY JUNQUEIRA


**Reviewed by**

Dr. Sara moeen

Lecturer Anatomy Department

DMC

# THANK YOU !